Meeting Wadoud - Raul - Sebastian.

Volatility coefficient:

$$\bar{b}^2(t,s) = \mathbb{E}\left[\|Pb\|^2(t,x) \mid P \cdot X = s \right]$$

$\bar{b}^2$ approximated by regression:

$$\bar{b}^2(t,s) = \underset{v \in V}{\arg\min}\ \mathbb{E}\left[\{ \|Pb\|^2(t,s) - v(t,s) \}^2 \mid PX = s \right]$$

Regression in time:

$$\bar{b}^2(t,s) = \arg\min \int_0^T \mathbb{E}\left[|Pb|^2(t,x) - v(t,Px) \right]^2 dt$$

Intuition Transport Map:

Paths: $X_\ell = X_{\ell-1} + R_{\ell,\ell-1} \Rightarrow PX_\ell = PX_{\ell-1} + PR_{\ell,\ell-1}$

$R_{\ell,\ell-1} \to 0$ as $\ell \to \infty$

$$\Rightarrow PX_\ell = (\Sigma + Q_{\ell,\ell-1}) PX_{\ell-1}$$

$$\Rightarrow PX_\ell = T_{\ell,\ell-1}(PX_{\ell-1})$$

Multilevel:

$$\hat{V}_1(PX_1) + \hat{V}_0(PX_0) \approx T_{0\to 1} PX_0$$

For this part look Amelie's slide # 57